

Oscar

Read more on the open source car project
– Oscar in the August 2009 issue of Digit

Car-Fi

Ford's new cars are going to be WiFi hot spots as
an extension to its SYNC systems



Tata's new Air Car

them in cooler climates and necessitating heating mechanisms like heat exchangers. While these cars are all set to hit the roads of France by 2011, it's their features that make you sit up and pay attention. Compressed Air cars usually use a motor powered by compressed air – or combine them with gasoline, diesel, electric or ethanol engines – to help gain higher speeds. France's MDI plans several such hybrids to tide over this bug.

Tata's vehicle, called the 'Minicat', is purely run on an air-compression engine, is a zero-emission vehicle and might even be launched this year, as the first batch has already completed production at its Uttarakhand factory in Pantnagar. The company claims that the vehicle can reach a speed of 96 kilometres per hour and cover a range of 300 kilometres on a full tank of air.

Pneumatic technologies

Cars like Toyota's Prius and Honda's Insight use a petrol engine and braking energy to generate on-board electricity which adds to the power of the vehicle. Engineers led by Prof. Hua Zhao at Brunel University, West London, England, have discovered another way of adapting a regular petrol engine to run on compressed air. The unique aspect of this technology is that the compressed air is generated by using the braking mechanism. Whenever the car brakes, the pistons compress air and push it into a compressed-air tank. If this comes through, it is sure to lead to greener, cheaper and healthier cars.

Automobile add-ons

There's much more happening on the automotive innovation front than flying, floating or greening-up. With current digital technology, nanotechnology and new materials, car designers seem to

have received a license to take their creative ideas into overdrive. Here are a few examples to make your jaw drop.

Shared personal cars

Led by Praveen Nahar, Coordinator of the Transportation and Automobile Design programme at NID, Ahmedabad, three students of the same institute have come up with three unique designs for the Royal Melbourne Institute of Technology (RMIT). These designs, which are going to be used by the Australian car-sharing firm, Flexicar. "Instead of owning a car, we might be sharing a car in the near future," says Prof. Nahar, referring to the concept behind these cars which entails that several drivers use a single car at different times on a regular basis.

These designs involve an incredible level of personalization: When you use a swipe card to enter the car, the car resets all your personal preferences to optimise your driving experience – and this includes the steering wheel, the seat arrangement and your favourite music.

The team includes three students, viz. Sameer Malkhare, Lalit Yaduvanshi and Gautam Nair. Prof. Nahar hopes that, in future, civic administrative bodies may even decide to allow such shared vehicles to receive priority on the streets, once they realise their benefits.

Chameleon Cars

Using a special treatment called Feint-Paint by the Spanish company Payola Forlids, BMW's MINI Chameleon changes colour with the weather. The paint on the car is treated with microscopic magnetic iron oxide particles, which, once a minor magnetic field passes through it, changes colour in a matter of seconds. The car is sensitive to both, temperature as well as moisture. This should be seen on the streets of Singapore sometime soon.

Robotic car companion

MIT and Audi have come up with the AIDA (Affective Intelligent Driving Agent) which is a robot head set up on your dashboard, with a 'neck' that allows it to swivel and laser colour projections on its surface to give it expressions. The



Riding high

AIDA is connected to sensors all over the interior and exterior of the car, which gives it information about the driver's state of mind (strength of the grip on the steering wheel, etc.), and the status of the seat-belt is off. It also helps the driver get to the destination quicker using alternative routes with the help of GPS.

Augmented Reality System

Imagine this: you're driving around a corner and you can't see what's coming round the bend; but your car shows you the entire scene from behind the wall, even before you've reached there.

Denso, a car part manufacturer in Japan, along with Yaser Sheikh of Carnegie Mellon University – and his team – have been researching a camera system that enables a driver to see through walls. While this is still in the nascent stage, the team hopes to put a video processor on the dashboard, which would receive a wireless signal from a roadside camera and project the scene from behind the wall directly on your windscreen, preventing the risk and bother of having to move your eyes from the road to a screen.

The road ahead

Of course, the question is, whether we'll be able to see any of these contraptions in our lifetimes. But then, did you ever dream you'd see battery powered scooters whizzing around silently on the streets of Indian cities? Hasn't GPS – once the stuff of science fiction – now become an integral part of thousands of cars in the developed world? Similarly, several of the technologies discussed above are already way beyond the concept stage and are already in use by people like you and me. Don't be surprised, then, to see your neighbour fly his car into his garage, one fine day. Who knows, you might even beat him to it. **d**